COMPUTER ENGINEERING

QUEZON CITY

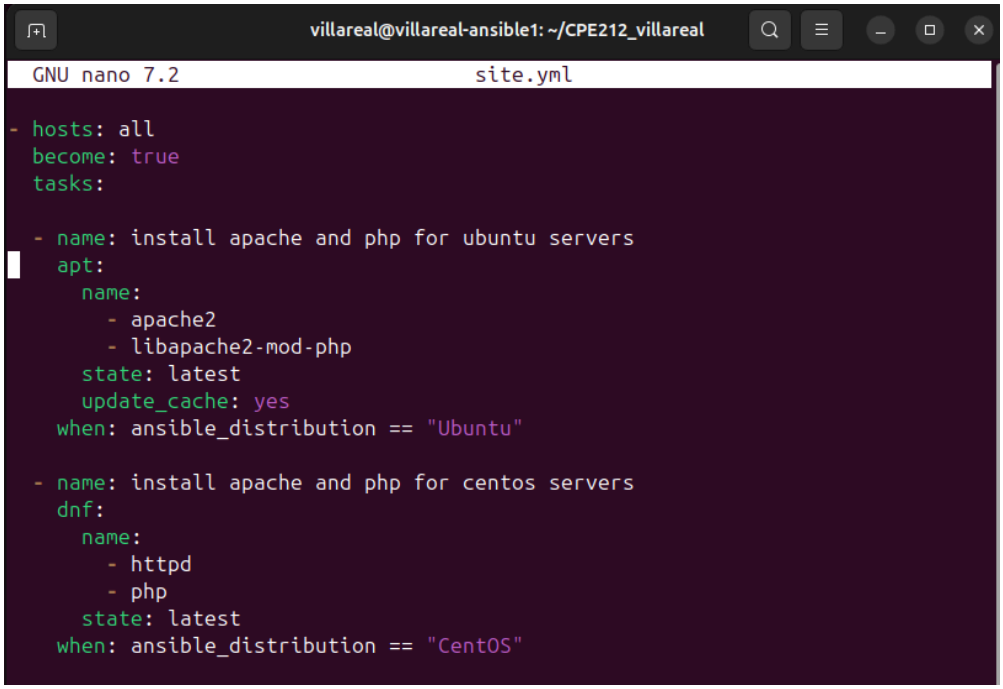
COURSE CODE/TITLE:	CPE 212 | Automating Server Management	**SECTION:**	CPE31S2
DATE PERFORMED:	9/3/26	**DATE SUBMITTED:**	9/3/26
NAME:	Villareal, Vanderohe Vernon	**INSTRUCTOR:**	Engr. Jimlord Quejado

ACTIVITY NO. 6.1 Targeting Specific Nodes

1. PROCEDURE OUTPUT

Task 1:

1. Create a new playbook and named it site.yml. Follow the commands as shown in the image below. Make sure to save the file and exit.



2. Edit the inventory file. Remove the variables we put in our last activity and group according to the image shown below:

```
villareal@villareal-ansible1: ~/CPE212_villareal
 GNU nano 7.2                          inventory
[web_servers]
192.168.56.109
192.168.56.108

[db_servers]
192.168.56.110

[file_servers]
192.168.56.110
```

3. Edit the site.yml by following the image below

```
villareal@villareal-ansible1: ~/CPE212_villareal
 GNU nano 7.2                          site.yml *
---

- hosts: all
  become: true
  pre_tasks:

  - name: install updates (centos)
    dnf:
      update_only: yes
      update_cache: yes
    when: ansible_distribution == "CentOS"

  - name: install updates (Ubuntu)
    apt:
      upgrade: dist
      update_cache: yes
    when: ansible_distribution == "Ubuntu"

- hosts: web_servers
  become: true
  tasks:

  - name: install apache and php for ubuntu servers
    apt:
      name:
        - apache2
        - libapache2-mod-php
      state: latest
```

```
      update_cache: yes
    when: ansible_distribution == "Ubuntu"

  - name: install apache and php for centos servers
    dnf:
      name:
        - httpd
        - php
      state: latest
    when: ansible_distribution == "CentOS"
```

Run the site.yml file and describe the result

```
villareal@villareal-ansible1:~/CPE212_villareal$ ansible-playbook -i inventory
 site.yml --ask-become-pass
BECOME password:

PLAY [all] *********************************************************************
**

TASK [Gathering Facts] *********************************************************
**
ok: [192.168.56.110]
ok: [192.168.56.108]
ok: [192.168.56.109]

TASK [install updates (centos)] ************************************************
**
skipping: [192.168.56.109]
skipping: [192.168.56.108]
ok: [192.168.56.110]

TASK [install updates (Ubuntu)] ************************************************
**
skipping: [192.168.56.110]
changed: [192.168.56.108]
changed: [192.168.56.109]

PLAY [web_servers] *************************************************************
**

TASK [Gathering Facts] *********************************************************
**
ok: [192.168.56.108]
ok: [192.168.56.109]

TASK [install apache and php for ubuntu servers] *******************************
**
ok: [192.168.56.109]
ok: [192.168.56.108]

TASK [install apache and php for centos servers] ********************************
**
skipping: [192.168.56.109]
skipping: [192.168.56.108]

PLAY RECAP *********************************************************************
**
192.168.56.108             : ok=4    changed=1    unreachable=0    failed=0
 skipped=2    rescued=0    ignored=0
192.168.56.109             : ok=4    changed=1    unreachable=0    failed=0
 skipped=2    rescued=0    ignored=0
192.168.56.110             : ok=2    changed=0    unreachable=0    failed=0
 skipped=1    rescued=0    ignored=0
```

- The results of running the playbook without issues.

4. Let's try to edit again the site.yml file. This time, we are going to add plays targeting the other servers. This time we target the db_servers by adding it on the current site.yml. Below is an example: (Note add this at the end of the playbooks from task 1.3.

```
villareal@villareal-ansible1: ~/CPE212_villareal
 GNU nano 7.2                          site.yml *
    name:
      - httpd
      - php
    state: latest
  when: ansible_distribution == "CentOS"

- hosts: db_servers
  become: true
  tasks:

  - name: install mariadb package (CentOS)
    yum:
      name: mariadb-server
      state: latest
    when: ansible_distribution == "CentOS"

  - name: "Mariadb- Resatrting/Enabling"
    service:
      name: mariadb
      state: restarted
      enabled: true

  - name: install mariadb package (Ubuntu)
    apt:
      name: maridb-server
      state: latest
    when: ansible_distribution == "Ubuntu"
```

Run the site.yml file and describe the result

```
villareal@villareal-ansible1: ~/CPE212_villareal
skipping: [192.168.56.109]
skipping: [192.168.56.108]

PLAY [db_servers] ***************************************************************
**

TASK [Gathering Facts] **********************************************************
**
ok: [192.168.56.110]

TASK [install mariadb package (CentOS)] *****************************************
**
changed: [192.168.56.110]

TASK [Mariadb- Resatrting/Enabling] *********************************************
**
changed: [192.168.56.110]

TASK [install mariadb package (Ubuntu)] *****************************************
**
skipping: [192.168.56.110]

PLAY RECAP **********************************************************************
**
192.168.56.108             : ok=4    changed=0    unreachable=0    failed=0
 skipped=2    rescued=0    ignored=0
192.168.56.109             : ok=4    changed=0    unreachable=0    failed=0
 skipped=2    rescued=0    ignored=0
192.168.56.110             : ok=5    changed=2    unreachable=0    failed=0
 skipped=2    rescued=0    ignored=0
```

- When running the playbook there are still no issues, only additional outputs since there are new additions to the .yml file.

5. Go to the remote server (Ubuntu) terminal that belongs to the db_servers group and check the status for mariadb installation using the command: systemctl status mariadb. Do this on the CentOS server also. Describe the output.

```
[villareal@localhost ~]$ systemctl status mariadb
● mariadb.service - MariaDB 10.5 database server
     Loaded: loaded (/usr/lib/systemd/system/mariadb.service; enabled; preset: >
     Active: active (running) since Thu 2026-09-03 11:40:42 PST; 4min 8s ago
       Docs: man:mariadbd(8)
             https://mariadb.com/kb/en/library/systemd/
    Process: 10288 ExecStartPre=/usr/libexec/mariadb-check-socket (code=exited,>
    Process: 10313 ExecStartPre=/usr/libexec/mariadb-prepare-db-dir mariadb.ser>
    Process: 10424 ExecStartPost=/usr/libexec/mariadb-check-upgrade (code=exite>
   Main PID: 10405 (mariadbd)
     Status: "Taking your SQL requests now..."
      Tasks: 8 (limit: 22978)
     Memory: 75.2M (peak: 106.3M)
        CPU: 1.285s
     CGroup: /system.slice/mariadb.service
             └─10405 /usr/libexec/mariadbd --basedir=/usr
```

- On the CentOS server the mariadb is running and active

```
villareal@villareal-ansible1:~$ systemctl status mariadb
Unit mariadb.service could not be found.
```

```
villareal@villareal-ansible1:~$ systemctl status mariadb
Unit mariadb.service could not be found.
```

- On the ubuntu server it says no service can be found since the playbook and inventory is not written to install this package for the IPs of these servers.

6. Edit the site.yml again. This time we will append the code to configure installation on the file_servers group. We can add the following on our file

```
- hosts: file_servers
  become: true
  tasks:

  - name: install samba package
    package:
      name: samba
      state: latest
```

```
TASK [install samba package] ***************************************************
**
changed: [192.168.56.110]

PLAY RECAP *********************************************************************
**
192.168.56.108             : ok=4    changed=0    unreachable=0    failed=0
 skipped=2    rescued=0    ignored=0
192.168.56.109             : ok=4    changed=0    unreachable=0    failed=0
 skipped=2    rescued=0    ignored=0
192.168.56.110             : ok=7    changed=2    unreachable=0    failed=0
 skipped=2    rescued=0    ignored=0
```

- After running still no issues, additional output where samba package is now installed on file_servers.

Task 2:

1. **Edit the site.yml file. Add tags to the playbook. After the name, we can place the tags: name_of_tag. This is an arbitrary command, which means you can use any name for a tag.**

```
- name: install updates (centos)
  tags: always
  dnf:
    update_only: yes
    update_cache: yes
  when: ansible_distribution == "CentOS"

- name: install updates (Ubuntu)
  tags: always
  apt:
    upgrade: dist
    update_cache: yes
  when: ansible_distribution == "Ubuntu"
```

```
- name: install apache and php for ubuntu servers
  tags: apache,apache2,ubuntu
  apt:
    name:
      - apache2
      - libapache2-mod-php
    state: latest
    update_cache: yes
  when: ansible_distribution == "Ubuntu"

- name: install apache and php for centos servers
  tags: apache,centos,httpd
  dnf:
    name:
      - httpd
      - php
    state: latest
  when: ansible_distribution == "CentOS"
```

```
tags: centos,db,mariadb
yum:
  name: mariadb-server
  state: latest
when: ansible_distribution == "CentOS"

name: "Mariadb- Resatrting/Enabling"
service:
  name: mariadb
  state: restarted
  enabled: true

name: install mariadb package (Ubuntu)
tags: db,mariadb,ubuntu
apt:
  name: maridb-server
  state: latest
when: ansible_distribution == "Ubuntu"

osts: file_servers
ecome: true
asks:

name: install samba package
tags: samba
package:
  name: samba
  state: latest
```

Run the site.yml file and describe the result.

```
villareal@villareal-ansible1:~/CPE212_villareal$ ansible-playbook -i inventor
 site.yml --ask-become-pass
BECOME password:

PLAY [all] *********************************************************************
**

TASK [Gathering Facts] *********************************************************
**
ok: [192.168.56.109]
ok: [192.168.56.108]
ok: [192.168.56.110]

TASK [install updates (centos)] ************************************************
**
skipping: [192.168.56.109]
skipping: [192.168.56.108]
ok: [192.168.56.110]

TASK [install updates (Ubuntu)] ************************************************
**
skipping: [192.168.56.110]
ok: [192.168.56.108]
ok: [192.168.56.109]

PLAY [web_servers] *************************************************************
**

TASK [Gathering Facts] *********************************************************
**
ok: [192.168.56.109]
```

```
ok: [192.168.56.108]

TASK [install apache and php for ubuntu servers] ***************************
**
ok: [192.168.56.109]
ok: [192.168.56.108]

TASK [install apache and php for centos servers] ***************************
**
skipping: [192.168.56.109]
skipping: [192.168.56.108]

PLAY [db_servers] **********************************************************
**

TASK [Gathering Facts] *****************************************************
**
ok: [192.168.56.110]

TASK [install mariadb package (CentOS)] ************************************
**
ok: [192.168.56.110]

TASK [Mariadb- Resatrting/Enabling] ****************************************
**
changed: [192.168.56.110]

TASK [install mariadb package (Ubuntu)] ************************************
**
skipping: [192.168.56.110]

PLAY [file_servers] ********************************************************
```

```
**

TASK [Gathering Facts] *****************************************************
**
ok: [192.168.56.110]

TASK [install samba package] ***********************************************
**
ok: [192.168.56.110]

PLAY RECAP *****************************************************************
**
192.168.56.108             : ok=4    changed=0    unreachable=0    failed=0
 skipped=2    rescued=0    ignored=0
192.168.56.109             : ok=4    changed=0    unreachable=0    failed=0
 skipped=2    rescued=0    ignored=0
192.168.56.110             : ok=7    changed=1    unreachable=0    failed=0
 skipped=2    rescued=0    ignored=0

villareal@villareal-ansible1:~/CPE212_villareal$
```

- Still no issues and additional outputs

2. On the local machine, try to issue the following commands and describe each result:
   2.1 ansible-playbook --list-tags site.yml

```
villareal@villareal-ansible1:~/CPE212_villareal$ ansible-playbook -i inventory
 site.yml --list-tags

playbook: site.yml

  play #1 (all): all    TAGS: []
      TASK TAGS: [always]

  play #2 (web_servers): web_servers    TAGS: []
      TASK TAGS: [apache, apache2, centos, httpd, ubuntu]

  play #3 (db_servers): db_servers      TAGS: []
      TASK TAGS: [centos, db, mariadb, ubuntu]

  play #4 (file_servers): file_servers  TAGS: []
      TASK TAGS: [samba]
```

- This command only outputs tags within the site.yml file

2.2 ansible-playbook --tags centos --ask-become-pass site.yml

```
**
skipping: [192.168.56.109]
skipping: [192.168.56.108]

PLAY [db_servers] ***********************************************************
**

TASK [Gathering Facts] ******************************************************
**
ok: [192.168.56.110]

TASK [install mariadb package (CentOS)] *************************************
**
ok: [192.168.56.110]

PLAY [file_servers] *********************************************************
**

TASK [Gathering Facts] ******************************************************
**
ok: [192.168.56.110]

PLAY RECAP ******************************************************************
**
192.168.56.108             : ok=3    changed=0    unreachable=0    failed=0
 skipped=2    rescued=0    ignored=0
192.168.56.109             : ok=3    changed=0    unreachable=0    failed=0
 skipped=2    rescued=0    ignored=0
192.168.56.110             : ok=5    changed=0    unreachable=0    failed=0
 skipped=1    rescued=0    ignored=0
```

- This command only runs tasks that has a centos tag within the site.yml file

2.3 ansible-playbook --tags db --ask-become-pass site.yml

```
PLAY [db_servers] ***********************************************************
**

TASK [Gathering Facts] ******************************************************
**
ok: [192.168.56.110]

TASK [install mariadb package (CentOS)] *************************************
**
ok: [192.168.56.110]

TASK [install mariadb package (Ubuntu)] *************************************
**
skipping: [192.168.56.110]

PLAY [file_servers] *********************************************************
**

TASK [Gathering Facts] ******************************************************
**
ok: [192.168.56.110]

PLAY RECAP ******************************************************************
**
192.168.56.108             : ok=3    changed=0    unreachable=0    failed=0
 skipped=1    rescued=0    ignored=0
192.168.56.109             : ok=3    changed=0    unreachable=0    failed=0
 skipped=1    rescued=0    ignored=0
192.168.56.110             : ok=5    changed=0    unreachable=0    failed=0
 skipped=2    rescued=0    ignored=0
```

- This command only runs tasks that has a db tag within the site.yml file

2.4 ansible-playbook --tags apache --ask-become-pass site.yml

```
TASK [install apache and php for centos servers] ****************************
**
skipping: [192.168.56.109]
skipping: [192.168.56.108]

PLAY [db_servers] ***********************************************************
**

TASK [Gathering Facts] ******************************************************
**
ok: [192.168.56.110]

PLAY [file_servers] *********************************************************
**

TASK [Gathering Facts] ******************************************************
**
ok: [192.168.56.110]

PLAY RECAP ******************************************************************
**
192.168.56.108             : ok=4    changed=0    unreachable=0    failed=0
 skipped=2    rescued=0    ignored=0
192.168.56.109             : ok=4    changed=0    unreachable=0    failed=0
 skipped=2    rescued=0    ignored=0
192.168.56.110             : ok=4    changed=0    unreachable=0    failed=0
 skipped=1    rescued=0    ignored=0
```

- This command only runs tasks that has a apache tag within the site.yml file

2.5 ansible-playbook --tags “apache,db” --ask-become-pass site.yml

```
PLAY [db_servers] ***********************************************************
**

TASK [Gathering Facts] ******************************************************
**
ok: [192.168.56.110]

TASK [install mariadb package (CentOS)] *************************************
**
ok: [192.168.56.110]

TASK [install mariadb package (Ubuntu)] *************************************
**
skipping: [192.168.56.110]

PLAY [file_servers] *********************************************************
**

TASK [Gathering Facts] ******************************************************
**
ok: [192.168.56.110]

PLAY RECAP ******************************************************************
**
192.168.56.108             : ok=4    changed=0    unreachable=0    failed=0
 skipped=2    rescued=0    ignored=0
192.168.56.109             : ok=4    changed=0    unreachable=0    failed=0
 skipped=2    rescued=0    ignored=0
192.168.56.110             : ok=5    changed=0    unreachable=0    failed=0
 skipped=2    rescued=0    ignored=0
```

- This command only runs tasks that has apache and db tag within the site.yml file

Task 3:

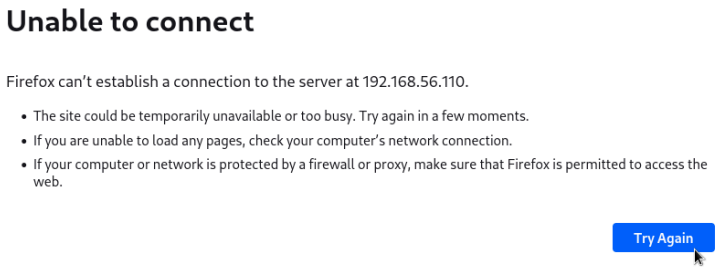
1. Edit the file site.yml and add a play that will automatically start the httpd on CentOS server.

```
- name: start httpd (CentOS)
  tags: apache,centos,httpd
  service:
    name: httpd
    state: latest
  when: ansible_distribution == "CentOS"
```

2. To test it, before you run the saved playbook, go to the CentOS server and stop the currently running httpd using the command sudo systemctl stop httpd. When prompted, enter the sudo password. After that, open the browser and enter the CentOS server's IP address. You should not be getting a display because we stopped the httpd service already.

```
[villareal@localhost ~]$ systemctl stop httpd
[villareal@localhost ~]$ systemctl status httpd
○ httpd.service - The Apache HTTP Server
     Loaded: loaded (/usr/lib/systemd/system/httpd.service; enabled; preset: d>
    Drop-In: /usr/lib/systemd/system/httpd.service.d
             └─php-fpm.conf
     Active: inactive (dead) since Thu 2026-09-03 14:10:02 PST; 9s ago
   Duration: 3h 37min 46.806s
       Docs: man:httpd.service(8)
    Process: 988 ExecStart=/usr/sbin/httpd $OPTIONS -DFOREGROUND (code=exited, >
   Main PID: 988 (code=exited, status=0/SUCCESS)
     Status: "Total requests: 0; Idle/Busy workers 100/0;Requests/sec: 0; Bytes>
        CPU: 21.349s
```





3. Go to the local machine and this time, run the site.yml file. Then after running the file, go again to the CentOS server and enter its IP address on the browser. Describe the result. To automatically enable the service every time we run the playbook, use the command enabled: true similar to Figure 7.1.2 and save the playbook

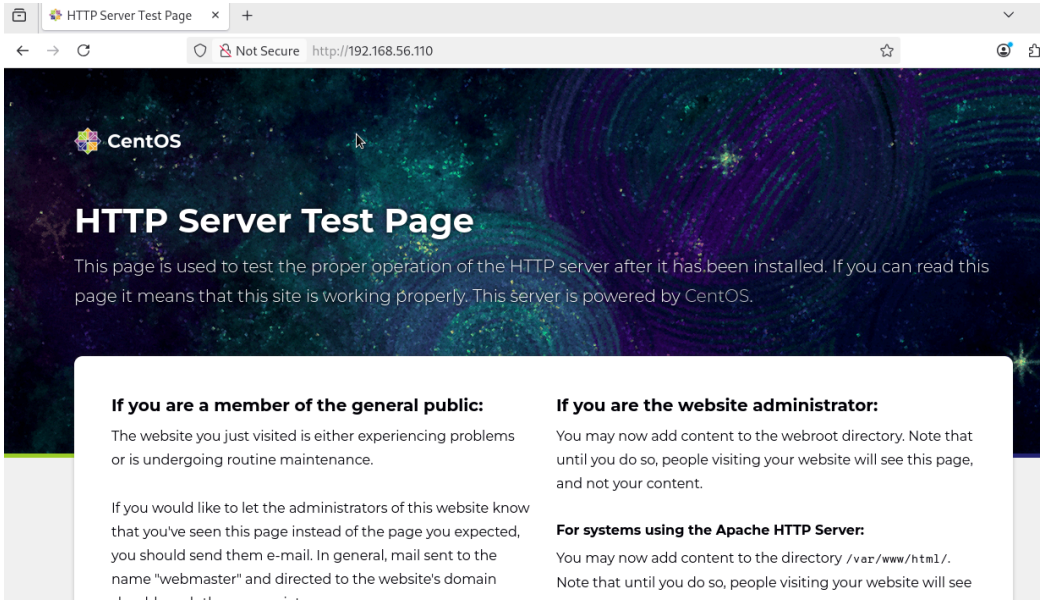
```
TASK [install mariadb package (Ubuntu)] ***********************************
***
skipping: [192.168.56.110]

PLAY [file_servers] *******************************************************
***

TASK [Gathering Facts] ****************************************************
***
ok: [192.168.56.110]

TASK [install samba package] **********************************************
***
ok: [192.168.56.110]

PLAY RECAP ****************************************************************
***
192.168.56.108             : ok=4    changed=0    unreachable=0    failed=0
  skipped=3    rescued=0    ignored=0
192.168.56.109             : ok=4    changed=0    unreachable=0    failed=0
  skipped=3    rescued=0    ignored=0
192.168.56.110             : ok=7    changed=1    unreachable=0    failed=0
  skipped=2    rescued=0    ignored=0
```



- After running the playbook the httpd on the centos server is running and active once again.

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

Supplementary Activity:

1. Creating a Role for Web Servers

```
villareal@villareal-ansible1: ~/CPE212_villareal
  GNU nano 7.2                                  web.yml
---
- name: configure web servers
  hosts: web_servers
  become: true
  roles:
    - webserver
```

2. Creating a Role for Web Servers

```
PLAY RECAP *********************************************************************
***
192.168.56.108             : ok=3    changed=0    unreachable=0    failed=0
  skipped=2    rescued=0    ignored=0
192.168.56.109             : ok=3    changed=0    unreachable=0    failed=0
  skipped=2    rescued=0    ignored=0
192.168.56.110             : ok=3    changed=0    unreachable=0    failed=0
  skipped=2    rescued=0    ignored=0
```

3. Creating a Role for Database Servers

```
villareal@villareal-ansible1: ~/CPE212_villareal
  GNU nano 7.2                                   db.yml
---
- name: configure databse servers
  hosts: db_servers
  become: true
  roles:
    - database
```

```
PLAY RECAP *********************************************************************
***
192.168.56.110             : ok=3    changed=0    unreachable=0    failed=0
  skipped=0    rescued=0    ignored=0
```

```
[villareal@localhost ~]$ systemctl status mariadb
● mariadb.service - MariaDB 10.5 database server
     Loaded: loaded (/usr/lib/systemd/system/mariadb.service; enabled; preset:
     Active: active (running) since Thu 2026-09-03 14:41:47 PST; 40min ago
       Docs: man:mariadbd(8)
             https://mariadb.com/kb/en/library/systemd/
    Process: 32382 ExecStartPre=/usr/libexec/mariadb-check-socket (code=exited,
    Process: 32404 ExecStartPre=/usr/libexec/mariadb-prepare-db-dir mariadb.ser
    Process: 32459 ExecStartPost=/usr/libexec/mariadb-check-upgrade (code=exite
   Main PID: 32439 (mariadbd)
     Status: "Taking your SQL requests now..."
      Tasks: 8 (limit: 22978)
     Memory: 73.3M (peak: 73.6M)
        CPU: 1.460s
     CGroup: /system.slice/mariadb.service
             └─32439 /usr/libexec/mariadbd --basedir=/usr
```

4. Combining Multiple Roles in a Playbook

```
villareal@villareal-ansible1: ~/CPE212_villareal
  GNU nano 7.2                          site.yml
---
- name: configure web servers
  hosts: web_servers
  become: true
  roles:
    - role: webserver
      tags:
        - web

- name: configure databse servers
  hosts: db_servers
  become: true
  roles:
    - role: database
      tags:
        - db
```

```
PLAY RECAP *********************************************************************
***
192.168.56.108             : ok=3    changed=0    unreachable=0    failed=0
  skipped=2    rescued=0    ignored=0
192.168.56.109             : ok=3    changed=0    unreachable=0    failed=0
  skipped=2    rescued=0    ignored=0
192.168.56.110             : ok=6    changed=0    unreachable=0    failed=0
  skipped=2    rescued=0    ignored=0
```

To orchestrate multiple roles in a single playbook for different server groups. Add tags to each role to allow selective execution, e.g., tags: web for webserver, tags: db for database.

ansible-playbook -i inventory site.yml --tags web --ask-become-pass

```
PLAY RECAP *********************************************************************
***
192.168.56.108             : ok=3    changed=0    unreachable=0    failed=0
  skipped=2    rescued=0    ignored=0
192.168.56.109             : ok=3    changed=0    unreachable=0    failed=0
  skipped=2    rescued=0    ignored=0
192.168.56.110             : ok=4    changed=0    unreachable=0    failed=0
  skipped=2    rescued=0    ignored=0
```

ansible-playbook -i inventory site.yml --tags db --ask-become-pass

```
PLAY RECAP *********************************************************************
***
192.168.56.108             : ok=1    changed=0    unreachable=0    failed=0
  skipped=0    rescued=0    ignored=0
192.168.56.109             : ok=1    changed=0    unreachable=0    failed=0
  skipped=0    rescued=0    ignored=0
192.168.56.110             : ok=4    changed=0    unreachable=0    failed=0
  skipped=0    rescued=0    ignored=0
```

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used: **ChatGPT**

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use: **Explanation of concepts,error solutions, and unfamiliar commands for supplementary**

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use: **AI is used to explain concepts like roles, webserver, and dbservers in the playbook as well as solving specific errors encountered like incorrect IP inputs in the inventory file, finally it is used to ask about unfamiliar commands for the supplementary activity.**

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

By submitting this activity, I affirm that my use of AI complies with T.I.P.'s AI usage policy and does not exceed the permitted limits for similarity, automated content generation, or academic assistance.

I understand that any false, incomplete, or misleading disclosure may be subject to investigation in accordance with due process and may result in sanctions for violation of existing T.I.P. policies.



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